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(1) Research Program/Project/Study Title:	
SENSORY EVALUATION FOR CASSAVA FRIES: A PRODUCT DEVELOPMENT	
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Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title:	Sensory Evaluation for Cassava Fries Enhanced with Sweet Potato	
Component Leader:	Joesan D. Borres, PhD-BM	
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Component Title:	Sensory Evaluation for Cassava Fries Enhanced with Squash	
Component Leader:	Joesan D. Borres, PhD-BM	
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Component Title:	Sensory Evaluation for Cassava Fries Enhanced with Yam	
Component Leader:	Joesan D. Borres, PhD-BM	
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Component Title:	Sensory Evaluation for Cassava Fries Enhanced with Carrots	
Component Leader:	Joesan D. Borres, PhD-BM	
Academic Rank:	Associate Professor III	
Component Title:	Sensory Evaluation for Cassava Fries Enhanced with Moringga	
Component Leader:	Joesan D. Borres, PhD-BM	
Academic Rank:	Associate Professor III	
Component Title:	Sensory Evaluation for Cassava Fries Enhanced with Beerots	
Component Leader:	Joesan D. Borres, PhD-BM	
Academic Rank:	Associate Professor III	
Component Title:	Shelf Life and Nutritional Analysis of Cassava Fries Products	
Component Leader:	Joesan D. Borres, PhD-BM	
Academic Rank:	Associate Professor III	
Component Title:	Cassava Fries Products: A Feasibility Study	
Component Leader:	Joesan D. Borres, PhD-BM	
Academic Rank:	Associate Professor III	
Component Title:	Cassava Fries Products: A Marketing Study	
Component Leader:	Joesan D. Borres, PhD-BM	
Academic Rank:	Associate Professor III	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research ☐ Developmental/Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC
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	Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	
Province:	Capiz
Region:	VI

(5) Classification Research: ☐ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity cassava farmers, sellers and consumers	(8) Discipline/Field of Specialization Business Management Social Science
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Abstract

The global demand for healthier food alternatives has led to innovations in the fast-food industry, particularly in developing sustainable options that align with local agricultural practices. This study aims to explore the potential of cassava as a healthier alternative to traditional potato fries. The primary objective was to evaluate the sensory attributes appearance, taste, texture, and general acceptability of cassava fries prepared using various recipes.

An experimental methodology with a Completely Randomized Design (CRD) was employed, involving five treatments replicated three times. The sensory evaluation was conducted by 55 evaluators using a five-point Hedonic scale. Data were analyzed using One-Way ANOVA to determine significant differences among treatments.

The study found that Treatment E, which used a higher proportion of cassava, consistently received the highest scores across all sensory attributes, indicating its superior acceptability. The significant differences observed in the attributes suggest that factors such as ingredient proportions and preparation methods critically influence the sensory quality of cassava fries.

The findings underscore the potential of cassava fries as a healthier, sustainable alternative to traditional French fries. This innovation not only caters to health-conscious consumers but also supports local agriculture, aligning with the United Nations Sustainable Development Goals. Future research may focus on optimizing these attributes and exploring diverse flavor profiles to broadened consumer appeal.

KEYWORDS:

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Cassava fries, sensory evaluation, healthier alternative, sustainable food, fast-food innovation

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(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

French fries are a popular and sustainable potato product globally, known for their widespread appeal and cultural adaptability. The global French fries market was valued at USD 16.60 billion in 2023 and is projected to grow to USD 26.56 billion by 2032, with a compound annual growth rate of 5.6% (Fortune Business Insights, 2024). However, there is a growing trend towards the healthier eating, prompting innovations in the fast-food industry, including alternatives to traditional deep-fried potatoes.

The fast-food industry has been exploring new flavors and ingredients to cater to diverse palates and health-conscious consumers. For example, in South Korea, fast-food chains have introduced Kimchi-flavored fries, combining local culinary traditions with the global appeal of French fries (Pieterse, 2024).

This study aligns with the United Nations Sustainable Development Goals, particularly SDG 12 (Responsible Consumption and Production). By using cassava, a locally sourced and sustainable food production and reduces reliance on imported potatoes. The study also contributes to SDG 3 (Good Health and Well-being), by offering a healthier alternative to traditional deep-fried foods, potentially reducing the consumption of unhealthy fats and supporting better dietary choices.

The research gap lies in the lack of comprehensive studies on the production and market potential of cassava-based fries, particularly in Capiz, where cassava production is significant. The Philippines is projected to produce 709.91 thousand metric tons of cassava in the second quarter of 2023, highlighting its availability as a raw material (PSA, 2024).

This study focuses on developing a new method of preparing cassava fries that are not only healthier but also align with local agricultural practices and culinary preferences. Prior art CN105053898A the invention discloses a making method of cassava whole flour chips. The making method of cassava whole flour chips comprises the following steps: pre-boiling cassava blocks, boiling, and processing to form mashed cassava; processing the mashed cassava through a noodle press and a knife to form cassava strips; and putting the cassava strips in an airfryer, coating with a layer of edible oil, frying at 180DEG C for 5min, overturning, and frying at 180Deg C for 5 min. Pre-boiling keeps the integrity of cells as possible; boiling and mashing realize homogeneity of the appearance color and the mouthfeel of the cassava chips; pressing of the mashed cassava by the noodle press makes the cassava chips have a relatively homogenous thickness; the method reduces the use of amount of the edible oil; and the method reduces the oil absorption amount of the cassava chips. The method is simple, greatly reduces the oil absorption rate of the cassava chips, and effectively reduces the over oxidation degree of oil in a novel heating mode; the novel heating mode makes the oil absorption rate of the cassava chips decrease to about 0.7% from 5.6%; and the edible oil is added several times, so multi-time over oxidation of oil at a high temperature is avoided (Zhang Zhenwen & Lin Liming, 2015). This present study builds on previous methods by incorporating additional ingredients to enhance the nutritional value and exploring different preparation processes to maintain the product's integrity and appeal.

Cassava locally known as balinghoy is a good source of vitamin C, thiamine, riboflavin, and niacin. Combining with cheese and dried parsley which are good source of vitamins A, D, K, zinc and antioxidants can optimize its potential. Since, the purpose for this research is to provide a healthier, sustainable alternative to traditional French fries that can cater to the growing health-conscious consumer market while supporting local agriculture. By infusing local flavors and using innovative cooking methods, this study aims to contribute to the global conversation on sustainable food production and consumption.

In general, this study determines the sensory evaluation of cassava fries in terms of appearance, taste, texture and general acceptability.

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Materials and Methods. The materials used in this study were potatoes, cassava, cornflour, cheese, iodized salt, pepper, dried parsley and coconut oil.The tools used in the prepatation were: measuring spoons, mixing bowl, flat tray, pipping bag, aluminum foil, frying pan, pot, basin, chopping board, strainer, knife, food grammar, food processor, and mixer.

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This is an experimental method in Completely Randomized Design (CRD) with five treatments, replicated three times. A sensory evaluation score sheet based on five-point Hedonic scale was used to determine the sensory evaluation of cassava fries in terms of appearance, taste, texture, and general acceptability.

Table 1

Scoring Guide for Sensory Evaluation of Cassava Fries

Score	Interval	Appearance	Taste	Texture	General Acceptability
5	4.21 – 5.00	Extremely Appealing	Extremely Delicious	Extremely Fine	Extremely Acceptable
4	3.41 – 4.20	Slightly Appealing	Slightly Delicious	Slightly Fine	Slightly Acceptable
3	2.61 – 3.40	Neither Appealing Nor Not Appealing	Neither Delicious Nor Not Delicious	Neither Fine Nor Not Fine	Neither Acceptable Nor Not Acceptable
2	1.81 – 2.60	Slightly Not Appealing	Slightly Not Delicious	Slightly Not Fine	Slightly Not Acceptable
1	1.00 – 1.80	Extremely Not Appealing	Extremely Not Delicious	Extremely Not Fine	Extremely Not Acceptable

The experimental lay-out using Completely Randomized Design (CRD) below shows the treatment and replication of the recipe. The order of CRD in this study, were determine by assigning a letter for treatments and number for replication. The randomization procedure for allotting the treatments to various units were determine by drawing of lots.

C ₃	B ₂	B ₁	A ₂	E ₃
D ₁	A ₃	E ₁	D ₃	A ₁
C ₂	D ₂	B ₃	C ₁	E ₂

Figure 1. Experimental layout of the study using Completely Randomized Design.

Procedures. Peel the cassava. Wash the cassava and put in a pot. Put water that enough to cover the cassava. Boil until it become soft. Drain and set aside to let it cool. Portion all dry ingredients and mixed, set aside. Portion the cube cassava. Cut the cassava into cubes. Remove the hard fiber like. Put it to the food processor. After which transfer it to the mixer and gradually add the dried ingredients. Then, put the mixture in a pipping bag. Cut the tip of the pipping bag. Pipe the cassava mixture in a flat tray about 2 inches long strip. After, cover with aluminum foil and put inside the refrigerator overnight.

In frying pan, heat coconut oil. Fry cassava fries in a low heat until it become crispy brown. Drain and it is ready to serve.

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Table 1

Distribution of Ingredients per Treatment of Cassava Fries

Ingredients	Treatment				
	A	B	C	D	E
Potato	500 g	350 g	250 g	150 g	
Cassava		150 g	250 g	350 g	500 g
Cheese	100 g	100 g	100 g	100 g	100 g
Cornflour	4 tbs	4 tbs	4 tbs	4 tbs	4 tbs
Iodized salt	½ tsp	½ tsp	½ tsp	½ tsp	½ tsp
Pepper	½ tsp	½ tsp	½ tsp	½ tsp	½ tsp
Dried parsley	¼ tsp	¼ tsp	¼ tsp	¼ tsp	¼ tsp
Yield	555 g	590 g	615 g	620 g	650 g

Data Gathering Procedures. The researcher asked permission from the Satellite College Director prior from the conduct of the study. The 55 panel of evaluators were chosen randomly through convenience sampling. The 5 faculty are composed of HM and FoodTech graduates. The 5 staff are randomly selected. All ten students are from BSHM Department. While, 50% of consumers composed of vendors and sellers of snackfood like fries, lumpia, kikiyam, etc, and the remaining composed of typical Pilareno consumers.

Most of the respondents in the study are classified as consumers 35 or 64%, with faculty and staff, and students each representing 10 or 18% of the total. Majority of the respondents are biological female comprising 43 or 78% and males only 12 or 22%. The 39 or 71% of the respondents belong to younger group (24 years old and below) and the remaining 16 or 29% are older group (25 years old and above). The results may heavily reflect the preferences and perceptions of younger, female consumers, which could be useful for businesses and researchers targeting this demographic. However, for broader applicability, it would be advisable to include a more balanced sample in terms of sex and age, ensuring that the findings are more representative of the general population.

The manner of sensory evaluation held in batches at Capiz State University – Pilar Campus last July 25, 2024. Evaluators were asked to drink water first before they taste each treatment. Informed Consent Form were also secured.

Table 2

Profile of the Respondents

Profile	f	%
Classification of the Panel of Evaluators		
Faculty and Staff	10	18
Students	10	18
Consumers	35	64
Total	55	100
Sex		
Male	12	22
Female	43	78
Total	55	100
Age		
Young (24 years old and below)	39	71
Old (25 years old and above)	16	29
Total	55	100

Data Analysis Procedures. After the sensory evaluation of the panel of evaluators, the sensory evaluation sheets were automatically gathered, checked if there is a missing answer, responses are tallied in the Excel and subjected to the following statistical analysis.

Mean was used to determine the sensory evaluation of cassava fries in terms of appearance, taste, texture, and general acceptability among five treatments.

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One-Way Analysis of Variance (ANOVA) was used to determine the significant difference of cassava fries in terms of appearance, taste, texture, and general acceptability among five treatments. Alpha level of significance was set at 0.01. Comparison between pairs of treatment means was tested using the Least Significance Differences (LSD).

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Sensory Evaluation of Cassava Fries in terms of Appearance, Taste, Texture, and General Acceptability among Five Treatments

Appearance. Treatment E received the highest mean score of 4.45, indicating it is extremely appealing to respondents. This was followed by Treatments B (3.91), D. (3.89), C (3.60), and A (2.56), with Treatment A being the least preferred.

Taste. Treatment E led with a mean score of 4.74, indicating it is extremely delicious. This was followed by Treatment D (4.21), C (4.13), B (3.93), and A (2.91).

Texture. Treatment E scored the highest with a mean of 4.95, indicating it is extremely fine, followed by Treatments C (4.29), D (4.27), B (4.01), and A (3.61).

General Acceptability. Consistent with other attributes, Treatment E had the highest mean score of 4.52 which means it is extremely acceptable. This was followed by Treatments D (4.15), C (3.98), B (3.73), and A (2.84).

In general, Treatment E consistently scores the highest across all sensory criteria, suggesting it is the most preferred treatment for cassava fries in terms of appearance, taste, texture, and general acceptability.

Table 3

Sensory Evaluation of Cassava Fries in terms of Appearance, Taste, Texture, and General Acceptability among Five Treatments

Sensory Evaluation	M	Verbal Description
<i>For Appearance</i>		
Treatment A	2.56	Slightly Not Appealing
Treatment B	3.91	Slightly Appealing
Treatment C	3.60	Slightly Appealing
Treatment D	3.89	Slightly Appealing
Treatment E	4.45	Extremely Appealing
<i>For Taste</i>		
Treatment A	2.91	Neither Delicious Nor Not Delicious
Treatment B	3.93	Slightly Delicious
Treatment C	4.13	Slightly Delicious
Treatment D	4.21	Extremely Delicious
Treatment E	4.74	Extremely Delicious
<i>For Texture</i>		
Treatment A	3.61	Slightly Fine
Treatment B	4.01	Slightly Fine
Treatment C	4.29	Extremely Fine
Treatment D	4.27	Extremely Fine
Treatment E	4.95	Extremely Fine
<i>For General Acceptability</i>		
Treatment A	2.84	Slightly Not Acceptable
Treatment B	3.73	Slightly Acceptable
Treatment C	3.98	Slightly Acceptable
Treatment D	4.15	Slightly Acceptable
Treatment E	4.52	Extremely Acceptable

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Significant Difference of Cassava Fries in terms of Appearance, Taste, Texture, and General Acceptability among Five Treatments

Appearance. There is a significant difference ($F\text{-value} = 79.39$, $p < 0.0001$) on appearance among five treatments. Therefore, the null hypothesis stated that there is no significant difference of cassava fries in terms of appearance is rejected.

Treatment E scored the highest (4.45), followed by B (3.91), D (3.89), C (3.60), and A (2.56). The significant $F\text{-value}$ and pairwise differences suggest that factors influencing the appearance, vary widely among the treatments. Treatments B and D are not significantly different from each other but are from the others, indicating similar visual appeal.

Taste. There is a significant difference ($F\text{-value} = 440.13$, $p < 0.0001$) in taste perception among the treatments. Therefore, the null hypothesis stated that there is no significant difference of cassava fries in terms of taste is rejected.

Treatment E leads with a mean score of 4.74, suggesting it had a more favorable flavor. Treatments C and D, with scores of 4.31 and 4.21, respectively, are not significant different from each other but differ significantly from A, B, and E.

Texture. There is a significant difference ($F\text{-value} = 145700.97$, $p < 0.0001$) in texture among the treatments. Therefore, the null hypothesis stated that there is no significant difference of cassava fries in terms of texture is rejected.

Treatment E scored the highest (4.95), indicating a desirable texture. Treatments B, C, and D, although close, showed significant differences from each other and from Treatments A and E.

General Acceptability. There is a significant difference ($F\text{-value} = 73.84$, $p < 0.0001$) on general acceptability among the treatments. Therefore, the null hypothesis stated that there is no significant difference of cassava in terms of general acceptability is rejected.

Treatment E's highest score (4.52) underscores its overall favorable reception, likely due to a balanced combination of appearance, taste, and texture.

Table 4

Significant Difference of Cassava Fries in terms of Appearance, Taste, Texture, and General Acceptability among Five Treatments

Source	DF	Sum of Square	Mean Square	F Value	Pr (> F)
For Appearance					
Treat	4	320.1285	80.0321	79.39	0.0000
Error	820	826.6667	1.0081		
Total	824	1146.7952			
For Taste					
Treat	4	885.6745	221.4186	440.13	0.0000
Error	2946	1482.0646	0.5031		
Total	2950	2367.7391			
For Texture					
Treat	4	46856.7696	11714.1924	145700.97	0.0000
Error	276234	22208.9002	0.0804		
Total	276238	69065.6698			
For General Acceptability					
Treat	4	262.2230	65.5558	73.84	0.0000
Error	820	727.9758	0.8878		
Total	824	990.1988			

Note: 0.01 alpha level of significance

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Table 4.a

Least Significant Difference (LSD) Test

Treat	means
For Appearance	
Treatment A	2.56 d
Treatment B	3.91 b
Treatment C	3.60 c
Treatment D	3.89 b
Treatment E	4.45 a
For Taste	
Treatment A	2.91 d
Treatment B	3.93 c
Treatment C	4.13 b
Treatment D	4.21 b
Treatment E	4.74 a
For Texture	
Treatment A	3.61 e
Treatment B	4.01 d
Treatment C	4.29 b
Treatment D	4.27 c
Treatment E	4.95 a
For General Acceptability	
Treatment A	2.84 d
Treatment B	3.73 c
Treatment C	3.98 b
Treatment D	4.15 b
Treatment E	4.52 a

Note: Means with the same letter are not significantly different

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Among the five treatments evaluated, Treatment E consistently received the highest scores across all attributes – appearance, taste, texture, and general acceptability. This indicates that Treatment E was the most preferred by respondents and suggests that the combination of preparation methods and ingredients used in this treatment led to acceptable product and more yield.

Significant differences were observed in the visual appeal of the cassava fries, with Treatment E being the most visually appealing. This highlights the importance of appearance in influencing consumer perception and acceptance, as first impressions can significantly impact the overall eating experience.

Taste emerged as a critical determinant of product acceptability, with Treatment E scoring the highest. The significant variation in taste ratings among the treatments suggests that factors such as seasoning and preparation techniques are crucial for enhancing the flavor profile of cassava fries.

The texture of the cassava fries, showed significant differences across treatments. Treatment E was rated the best in texture, indicating an ideal balance of crispiness and tenderness. This underscores the importance of achieving the right texture in fried products, as it greatly influences consumer satisfaction.

The general acceptability scores, which integrate all sensory attributes, further confirmed the superior quality of Treatment E. The study highlights the importance of optimizing sensory attributes such as appearance, taste, and texture to enhance the overall acceptability of food products. The findings provide valuable insights for researcher seeking to improve or develop new cassava fries products, emphasizing the need to focus on these key attributes to meet consumer preferences.

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Recommendations

Treatment E consistently scored highest in appearance, taste, texture, and general acceptability. It is recommended to replicate the preparation methods and ingredients used in Treatment E across other production batches to ensure consistency and maintain high-quality standards.

Further study for shelf life and nutrient analysis of cassava fries to know its macronutrients, micronutrients, and anti-nutritional facts to attract health conscious individuals.

Further explore and refine the specific ingredients (ex. type of oil, seasoning) and cooking methods (ex. frying temperature, time) that contributed to Treatment E's success. These factors may be standardized and optimized to consistently achieve the desired product quality.

Based on the sensory analysis, further research for developing new variants of cassava fries that cater to diverse tastes and preferences. For example, offering options with different flavors or textures can appeal to a broader audience.

Use the demographic insights to tailor marketing strategies. With a predominant young and female consumer base, marketing campaigns could emphasize health benefits, convenience, and trendy flavors that resonate with these groups.

Considering the positive reception of certain treatments, explore expanding the product to new markets, such as schools, cafes, or health food stores, where the appeal of a unique, nutritious snack could be strong.

Invest in attractive, convenient, and sustainable packaging solutions that appeal to on-the-go consumers and enhance the overall product experience.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Cassava fries

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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